

KARTIKEYA VEMULA

Analytics Engineer | dbt | BigQuery | GCP Certified | Power BI (PL-300)

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Open to relocate across India

PROFILE

Designs trusted analytics systems by combining warehouse engineering, automated testing, cloud infrastructure, and decision-ready reporting. GCP-certified (Professional Data Engineer, Associate Cloud Engineer) and Microsoft PL-300 certified, with experience spanning dimensional modeling, dbt transformation layers, cloud warehouses, and stakeholder-facing BI.

CERTIFICATIONS

Google Cloud Certified: Professional Data Engineer

Google Cloud Certified: Associate Cloud Engineer

Microsoft PL-300: Power BI Data Analyst Associate

Kaggle Advanced SQL

dbt Fundamentals

Snowflake Data Warehousing Workshop

Cisco Data Analytics Essentials

Google Analytics Certification

CORE SKILLS

Analytics Engineering: dbt Core, Dimensional Modeling, Star Schema, ELT/ETL Pipelines, Data Quality Testing, Incremental Models, Cross-Database Portability, Data Contracts, Semantic Layer & Data Mart Design, Automated Testing, Data Observability

Cloud & Warehousing: Google Cloud Platform, BigQuery, Snowflake, PostgreSQL, Docker, GitHub Actions (Scheduled Orchestration)

SQL & Python: SQL (Window Functions, CTEs, Query Optimization), Python (pandas, NumPy, SciPy, scikit-learn, XGBoost), Data Validation, Root Cause Analysis, Anomaly Detection

BI & Visualization: Power BI (PL-300), DAX, Power Query, Plotly Dash, Streamlit, Looker Studio, KPI Reporting, Executive Reporting

Engineering Practices: CI/CD for Data Pipelines, Version Control (Git), Reproducibility, Technical Documentation, Stakeholder Communication, Cross-Functional Collaboration, Statistical Analysis, A/B Testing, Product Analytics

EXPERIENCE

Graduate Research Assistant, Risk Analytics — UMBC, Baltimore

Oct 2025 – Jan 2026

Project: Extreme Value AQI Risk Modeling, Baltimore

- Built an **EPA AirData ETL pipeline** converting raw air-quality feeds into a validated **6,573-record dataset (2006–2023)** spanning AQI, pollutant, and meteorological variables — the single source the downstream risk model was allowed to trust.
- Designed structured **data quality checks** across missing values, outliers, and station-level reporting inconsistencies, so anomalies were resolved before modeling rather than explained away afterwards.
- Modeled pollutant risk with **XGBoost (RMSE 8.99, R² 0.852)** to establish a defensible baseline for expected AQI behavior.
- Applied **Extreme Value Theory (Generalized Pareto Distribution)** across 98 exceedance events, quantifying a **6.65% conditional tail-risk probability** for extreme pollution episodes.
- Translated model outputs into **stakeholder-ready risk reporting** that framed exceedance likelihood and tail-risk in language a non-technical monitoring team could act on directly.

Machine Learning Research Assistant — National Institute of Technology Puducherry

May 2023 – Jul 2023

Karaikal, India · Project: Fish Detection & Recognition System

- Engineered an end-to-end **computer vision pipeline in TensorFlow/Keras** classifying four fish species, structuring ingestion, augmentation, training, and evaluation as a single **reproducible workflow**.
- Standardized a **572-image labeled dataset to 143 images per class** through augmentation, removing the class imbalance that would otherwise have made accuracy a misleading success metric.
- Established a reproducible training pipeline using **transfer learning with pre-trained CNN architectures**, improving generalization on a small dataset and controlling overfitting.
- Architected the codebase as **reusable, independently runnable notebooks** separating data preparation from model training, so either stage could be re-run without repeating the other.
- Documented datasets, repository structure, and workflow so another contributor could **reproduce the complete training pipeline** without rebuilding the project.

PROJECTS

Instacart Intelligence Platform

Python, PostgreSQL, dbt Core, SQL, Docker, GitHub Actions, BigQuery, Power BI, DAX

Built an end-to-end analytics engineering platform on 3.4M+ orders and 33.8M order-item rows — from warehouse ingestion through tested dbt marts to an executive-ready Power BI dashboard.

- Engineered a dimensional warehouse over **3,421,083 orders and 33,819,106 order-item rows** (206,209 customers, 49,688 products), staging raw extracts through a star schema into purpose-built marts for lifecycle, retention, basket affinity, and category performance.
- Hardened the transformation layer into a production-ready system with **19 dbt models and 238 automated data tests** passing with zero errors, including singular tests that assert time-grid completeness and reconcile mart totals back to order volume — so a dashboard figure cannot silently drift from the fact table.

- Authored every model with **dbt cross-database macros** rather than engine-specific SQL, allowing the same transformation logic to target **PostgreSQL and BigQuery**, and containerized the project with Docker plus a scheduled GitHub Actions build.
- **Pre-aggregated all reporting marts into reusable analytical data products in dbt**, consumed independently by both the Power BI dashboard and a Plotly Dash app, while the 33.8M-row fact stays in the warehouse so the BI layer never recomputes aggregates on refresh.
- Segmented 206,209 customers into four behavioral cohorts — **69,635 Moderate Engagement, 62,818 Loyal Routine, 44,873 At Risk, 28,883 High Engagement** — giving lifecycle marketing a defined, targetable at-risk population, and documented the recency proxy used in place of true RFM given the dataset carries no timestamps or prices.
- Quantified order-sequence retention showing continuation falling to **53.70% by order 10**, converting a vague churn concern into a specific milestone for reorder nudges.
- Surfaced merchandising signal against a **59.01% platform reorder benchmark** — Produce as the leading department, **Banana the top SKU at 491,291 order items and an 84.51% reorder rate** — and reduced 1.2B theoretical product pairs to **254,490 statistically supported cross-sell candidates** using support, confidence, and lift thresholds.
- **Delivered an interactive Power BI dashboard** over the validated marts, with a documented DAX measure set and conditional-formatting rules that surface at-risk customers and low-reorder categories rather than totals alone.

KPI Reliability & Diagnostic Engine

Python, PostgreSQL, dbt Core, SQL, Streamlit, GitHub Actions, Anomaly Detection, Root Cause Analysis

- Built a KPI diagnostic engine monitoring **121,500 metric rows across 5 KPIs and 5 business dimensions** over 150 days, producing incident records carrying severity, confidence, and an attributed root-cause segment.
- Benchmarked static-threshold, rolling Z-score, and regression-residual detectors against labeled ground truth, then selected residual detection on evidence: **false positives fell from 2,958 to 563 (-81%)** and **F1 improved from 0.1878 to 0.5485** while **recall held at 100%**, meaning noise dropped without a single missed incident.
- **Engineered contribution-share root cause analysis** that ranks dimension combinations by their share of a KPI's degradation — isolating **APAC + Mobile + Safari as 100% of a 44.6% conversion-rate shortfall**, and **North America + Desktop + Chrome as 100% of a latency spike** — so triage starts at the responsible segment instead of a dashboard-wide search.
- **Implemented a 9-check data quality gate** (schema, types, nulls, duplicates, volume, freshness) scoring 0–100 and blocking inference on failure, so the engine never diagnoses a metric using data it cannot vouch for.
- Extended the platform with a **live operations track** ingesting the public Wikimedia Pageviews API on a **daily GitHub Actions schedule**, proving the detector runs unattended on external data nobody controls — and kept accuracy metrics scoped to the labeled dataset, since unlabeled production data cannot support them.
- Backed the codebase with **74 automated tests and byte-identical reproducible outputs**, so a reviewer re-running the pipeline lands on the same published figures.
- Shipped operational **Streamlit dashboards** and machine-readable **JSON/Markdown/HTML incident reports**, so engineering and business stakeholders could investigate the same incident from one shared source of truth.

Customer Intelligence Data Warehouse

Python, PostgreSQL, Docker, dbt Core, SQL, Star Schema, Power BI, DAX, Power Query

Built a customer analytics warehouse supporting retention, revenue concentration, and customer-value reporting.

- Converted noisy retail transaction data into **14 tested dbt models (41 passing tests)** governing **£3,279,364.61 in revenue across 5,047 customers**.
- Modeled repeat-purchase and customer-value intelligence, establishing **3,394 repeat buyers at a 67.25% repeat rate** and a single top customer worth **£62,131.42** — the concentration figures that shaped retention priorities.
- Developed country revenue marts exposing UK concentration at **£2,782,932.74 (84.86% of total)**, quantifying geographic dependency for performance monitoring.
- **Added product return-audit logic** separating genuine merchandise returns from operational transaction codes, preventing fees and adjustments from distorting product-level decisions.
- **Delivered a published interactive Power BI dashboard** unifying executive KPIs, repeat purchase, customer value, and country revenue.

EDUCATION

M.S. Data Science — University of Maryland, Baltimore County (UMBC), USA

2024 – 2026

QS Ranked, R1 Research University | GPA: 3.78 / 4.0 | Focus: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis

B.Tech Computer Science, AI & ML Specialization — GITAM University, India

2020 – 2024

8.89 GPA, First Class with Distinction

EXTRA-CURRICULAR & LEADERSHIP

- **Mentored peers** across Mathematics, Digital Logic, Data Communication, Design Thinking, and Introduction to Data Analysis & Machine Learning.
- **Led a 3-part UMBC peer-teaching series** on machine learning terminology, model training, technical explanation, and real-world case studies.
- **Supported chess club event coordination** and drafted social media content for the student organizing team.